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Name	Last Modified
Day_10_Task_5.ipynb	3 days ago
Day11_Task_2.ipynb	yesterday
Day11_Task_5.ipynb	yesterday
Day11_Task_6.ipynb	yesterday
ecommerce_cleaned.csv	yesterday
ecommerce_dirty_sample.csv	2 days ago
ecommerce_pandas_practice.ipynb	now
sample_student_performance.csv	3 days ago

ecommerce_pandas_practice

Code | Python 3 (ipykernel)

E-commerce Pandas Practice ¶

[1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv("ecommerce_dirty_sample.csv")
df.head()
```

[1]:

	order_id	order_date	customer_id	product_name	category	quantity	unit_price	total_amount	payment_method	order_status	city
0	ORD1001	2026-01-03	CUST001	Laptop	Electronics	1	899.99	899.99	Credit Card	Shipped	Chennai
1	ORD1002	2026-01-04	CUST002	wireless mouse	Electronics	2	19.99	39.98	UPI	Delivered	Bengaluru
2	ORD1003	2026-01-05	CUST003	Office Chair	Furniture	1	149.50	149.50	Cash on Delivery	Delivered	Mumbai
3	ORD1004	2026-01-06	CUST004	Smartphone	Electronics	1	699.00	699.00	NaN	Processing	Hyderabad
4	ORD1005	2026-01-07	CUST005	Coffee Maker	Home Appliances	2	45.00	90.00	UPI	Delivered	Pune

[2]:
```
df.tail()
```

[2]:

	order_id	order_date	customer_id	product_name	category	quantity	unit_price	total_amount	payment_method	order_status	city
99	ORD1100	2026-04-12	CUST100	Water Bottle	Home & Kitchen	4	12.00	48.00	UPI	Delivered	Ahmedabad
100	ORD1002	2026-01-04	CUST002	wireless mouse	Electronics	2	19.99	39.98	UPI	Delivered	Bengaluru
101	ORD1009	not-a-date	CUST009	Desk Lamp	Home Decor	2	24.50	49.00	Debit Card	Delivered	Jaipur
102	ORD1016	2026-01-18	CUST016	laptop	Electronics	1	899.99	899.99	credit card	Shipped	Chennai
103	ORD1034	2026-02-05	CUST034	Monitor	Electronics	1	699.00	699.00	Credit Card	Processing	Hyderabad

[3]:
```
df.shape
```

[3]:
```
(104, 11)
```

[4]:
```
df.columns
```

[4]:
```
Index(['order_id', 'order_date', 'customer_id', 'product_name', 'category',
       'quantity', 'unit_price', 'total_amount', 'payment_method',
       'order_status', 'city'],
      dtype='str')
```

[5]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 104 entries, 0 to 103
Data columns (total 11 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   order_id        104 non-null    str
 1   order_date      103 non-null    str
 2   customer_id     101 non-null    str
 3   product_name    103 non-null    str
 4   category        103 non-null    str
 5   quantity        104 non-null    str
 6   unit_price      102 non-null    float64
 7   total_amount    102 non-null    float64
 8   payment_method  103 non-null    str
 9   order_status    103 non-null    str
 10  city            103 non-null    str
dtypes: float64(2), str(9)
memory usage: 9.1 KB
```

[6]:
```
df.describe(include='all')
```

[6]:

	order_id	order_date	customer_id	product_name	category	quantity	unit_price	total_amount	payment_method	order_status	city
count	104	103	101	103	103	104	102.000000	1.020000e+02	103	103	103
unique	100	99	97	20	10	10	NaN	NaN	6	6	14
top	ORD1002	2026-01-04	CUST002	Desk Lamp	Electronics	1	NaN	NaN	UPI	Delivered	Chennai
freq	2	2	2	8	41	52	NaN	NaN	36	56	26
mean	NaN	NaN	NaN	NaN	NaN	NaN	9976.214706	9.824241e+04	NaN	NaN	NaN
std	NaN	NaN	NaN	NaN	NaN	NaN	98997.900387	9.901273e+05	NaN	NaN	NaN
min	NaN	NaN	NaN	NaN	NaN	NaN	-45.000000	0.000000e+00	NaN	NaN	NaN
25%	NaN	NaN	NaN	NaN	NaN	NaN	22.500000	4.800000e+01	NaN	NaN	NaN
50%	NaN	NaN	NaN	NaN	NaN	NaN	59.990000	8.997000e+01	NaN	NaN	NaN
75%	NaN	NaN	NaN	NaN	NaN	NaN	149.500000	1.495000e+02	NaN	NaN	NaN
max	NaN	NaN	NaN	NaN	NaN	NaN	999999.990000	1.000000e+07	NaN	NaN	NaN

[7]:
```
df['order_id'].nunique()
```

[7]:
```
100
```

[8]:
```
df.isna().sum()
```

[8]:
```
order_id          0
order_date        1
customer_id       3
product_name      1
category          1
quantity          0
unit_price        2
total_amount      2
payment_method    1
order_status      1
city              1
dtype: int64
```

[9]:
```
df.isna().sum()
```

[9]:
```
order_id          0
order_date        1
customer_id       3
product_name      1
category          1
quantity          0
unit_price        2
total_amount      2
payment_method    1
```

```
order_status      1
city              1
dtype: int64
```

[10]:
```
df.duplicated().sum()
```

[10]:
```
np.int64(3)
```

[11]:
```
df[df.duplicated(keep=False)]
```

[11]:

	order_id	order_date	customer_id	product_name	category	quantity	unit_price	total_amount	payment_method	order_status	city
1	ORD1002	2026-01-04	CUST002	wireless mouse	Electronics	2	19.99	39.98	UPI	Delivered	Bengaluru
8	ORD1009	not-a-date	CUST009	Desk Lamp	Home Decor	2	24.50	49.00	Debit Card	Delivered	Jaipur
15	ORD1016	2026-01-18	CUST016	laptop	Electronics	1	899.99	899.99	credit card	Shipped	Chennai
100	ORD1002	2026-01-04	CUST002	wireless mouse	Electronics	2	19.99	39.98	UPI	Delivered	Bengaluru
101	ORD1009	not-a-date	CUST009	Desk Lamp	Home Decor	2	24.50	49.00	Debit Card	Delivered	Jaipur
102	ORD1016	2026-01-18	CUST016	laptop	Electronics	1	899.99	899.99	credit card	Shipped	Chennai

[12]:
```
df = df.drop_duplicates().copy()
```

[13]:
```
df = df.rename(columns={
    "order_id": "OrderID",
    "order_date": "Date",
    "customer_id": "CustomerID",
    "product_name": "Product",
    "category": "Category",
    "quantity": "Units",
    "unit_price": "UnitPrice",
    "total_amount": "ReportedAmount",
    "payment_method": "PaymentMethod",
    "order_status": "OrderStatus",
    "city": "Region"
})
df.head()
```

[13]:

	OrderID	Date	CustomerID	Product	Category	Units	UnitPrice	ReportedAmount	PaymentMethod	OrderStatus	Region
0	ORD1001	2026-01-03	CUST001	Laptop	Electronics	1	899.99	899.99	Credit Card	Shipped	Chennai
1	ORD1002	2026-01-04	CUST002	wireless mouse	Electronics	2	19.99	39.98	UPI	Delivered	Bengaluru
2	ORD1003	2026-01-05	CUST003	Office Chair	Furniture	1	149.50	149.50	Cash on Delivery	Delivered	Mumbai
3	ORD1004	2026-01-06	CUST004	Smartphone	Electronics	1	699.00	699.00	NaN	Processing	Hyderabad
4	ORD1005	2026-01-07	CUST005	Coffee Maker	Home Appliances	2	45.00	90.00	UPI	Delivered	Pune

[14]:
```
for col in ["OrderID", "CustomerID", "Product", "Category", "PaymentMethod", "OrderStatus", "Region"]:
    df[col] = df[col].astype("string").str.strip()

df["Product"] = df["Product"].str.title()
df["Category"] = df["Category"].str.title()
df["PaymentMethod"] = df["PaymentMethod"].str.title()
df["OrderStatus"] = df["OrderStatus"].str.title()
df["Region"] = df["Region"].str.title()

df["Units"] = pd.to_numeric(df["Units"], errors="coerce")
df["UnitPrice"] = pd.to_numeric(df["UnitPrice"], errors="coerce")
df["ReportedAmount"] = pd.to_numeric(df["ReportedAmount"], errors="coerce")
df[["Units", "UnitPrice", "ReportedAmount"]].isna().sum()
```

[14]:
```
Units             2
UnitPrice         2
ReportedAmount    2
dtype: int64
```

[15]:
```
df[(df['Units'].isna()) | (df['Units'] <= 0) | (df['Units'] > 100)]
```

[15]:

	OrderID	Date	CustomerID	Product	Category	Units	UnitPrice	ReportedAmount	PaymentMethod	OrderStatus	Region
16	ORD1017	2026-01-19	CUST017	Wireless Mouse	Electronics	NaN	19.99	39.98	Upi	Delivered	Bengaluru
39	ORD1040	2026-02-11	CUST040	Water Bottle	Home & Kitchen	-2.0	12.00	48.00	Upi	Delivered	Ahmedabad
60	ORD1061	2026-03-04	CUST061	Laptop	Electronics	10000.0	899.99	899.99	Credit Card	Shipped	Chennai
95	ORD1096	2026-04-08	CUST096	Running Shoes	Fashion	NaN	79.99	79.99	Debit Card	Shipped	Delhi

[16]:
```
df[(df['UnitPrice'].isna()) | (df['UnitPrice'] <= 0) | (df['UnitPrice'] > 100000)]
```

[16]:

	OrderID	Date	CustomerID	Product	Category	Units	UnitPrice	ReportedAmount	PaymentMethod	OrderStatus	Region
11	ORD1012	2026-01-14	CUST012	Notebook Set	Stationery	5.0	NaN	24.95	Cash On Delivery	Delivered	Lucknow
18	ORD1019	2026-01-21	CUST019	Smartphone	Electronics	1.0	NaN	699.00	Upi	Processing	Hyderabad
35	ORD1036	2026-02-07	CUST036	Running Shoes	Fashion	1.0	999999.99	79.99	Debit Card	Shipped	Delhi
64	ORD1065	2026-03-08	CUST065	Coffee Maker	Home Appliances	2.0	-45.00	90.00	Upi	Delivered	Pune

[17]:
```
df[df['ReportedAmount'] < 0]
```

[17]:

	OrderID	Date	CustomerID	Product	Category	Units	UnitPrice	ReportedAmount	PaymentMethod	OrderStatus	Region

[18]:
```
df["Date"] = pd.to_datetime(df["Date"], errors="coerce", format="mixed")
df["Date"].isna().sum()
```

[18]:
```
np.int64(3)
```

[19]:
```
df[df['Date'].isna()]
```

[19]:

	OrderID	Date	CustomerID	Product	Category	Units	UnitPrice	ReportedAmount	PaymentMethod	OrderStatus	Region
8	ORD1009	NaT	CUST009	Desk Lamp	Home Decor	2.0	24.50	49.00	Debit Card	Delivered	Jaipur
45	ORD1046	NaT	CUST046	Laptop	Electronics	1.0	899.99	899.99	Credit Card	Shipped	Chennai
90	ORD1091	NaT	CUST091	Laptop	Electronics	1.0	899.99	899.99	Credit Card	Shipped	Chennai

[20]:
```
df["Year"] = df["Date"].dt.year
df["MonthNumber"] = df["Date"].dt.month
df["Day"] = df["Date"].dt.day
df["Month"] = df["Date"].dt.strftime("%Y-%m")
df = df.sort_values("Date", na_position="last").reset_index(drop=True)
df[["Date", "Year", "MonthNumber", "Day", "Month"]].head()
```

[20]:

	Date	Year	MonthNumber	Day	Month
0	2026-01-03	2026.0	1.0	3.0	2026-01
1	2026-01-04	2026.0	1.0	4.0	2026-01

2	2026-01-05	2026.0	1.0	5.0	2026-01
3	2026-01-06	2026.0	1.0	6.0	2026-01
4	2026-01-07	2026.0	1.0	7.0	2026-01

[21]:
```
df["DiscountPct"] = 0.0
valid_revenue_inputs = (
    df["Units"].notna()
    & df["UnitPrice"].notna()
    & (df["Units"] > 0)
    & (df["Units"] <= 100)
    & (df["UnitPrice"] > 0)
    & (df["UnitPrice"] <= 100000)
)
df["Revenue"] = np.where(
    valid_revenue_inputs,
    df["Units"] * df["UnitPrice"] * (1 - df["DiscountPct"]),
    np.nan
)
df[["OrderID", "Units", "UnitPrice", "DiscountPct", "Revenue"]].head()
```

[21]:

	OrderID	Units	UnitPrice	DiscountPct	Revenue
0	ORD1001	1.0	899.99	0.0	899.99
1	ORD1002	2.0	19.99	0.0	39.98
2	ORD1003	1.0	149.50	0.0	149.50
3	ORD1004	1.0	699.00	0.0	699.00
4	ORD1005	2.0	45.00	0.0	90.00

[22]:
```
df['Revenue'].isna().sum()
```

[22]:
```
np.int64(8)
```

[23]:
```
df.sort_values('Revenue', ascending=False).head(5)
```

[23]:

	OrderID	Date	CustomerID	Product	Category	Units	UnitPrice	ReportedAmount	PaymentMethod	OrderStatus	Region	Year	MonthNumber	Day	Mont
0	ORD1001	2026-01-03	CUST001	Laptop	Electronics	1.0	899.99	899.99	Credit Card	Shipped	Chennai	2026.0	1.0	3.0	2026
13	ORD1016	2026-01-18	CUST016	Laptop	Electronics	1.0	899.99	899.99	Credit Card	Shipped	Chennai	2026.0	1.0	18.0	2026
28	ORD1031	2026-02-02	<NA>	Laptop	Electronics	1.0	899.99	899.99	Credit Card	Shipped	Chennai	2026.0	2.0	2.0	2026
99	ORD1046	NaT	CUST046	Laptop	Electronics	1.0	899.99	899.99	Credit Card	Shipped	Chennai	NaN	NaN	NaN	Na
100	ORD1091	NaT	CUST091	Laptop	Electronics	1.0	899.99	899.99	Credit Card	Shipped	Chennai	NaN	NaN	NaN	Na

[24]:
```
df[['OrderID', 'Date', 'Region', 'Product', 'Units', 'UnitPrice', 'DiscountPct', 'Revenue']].head()
```

[24]:

	OrderID	Date	Region	Product	Units	UnitPrice	DiscountPct	Revenue
0	ORD1001	2026-01-03	Chennai	Laptop	1.0	899.99	0.0	899.99
1	ORD1002	2026-01-04	Bengaluru	Wireless Mouse	2.0	19.99	0.0	39.98
2	ORD1003	2026-01-05	Mumbai	Office Chair	1.0	149.50	0.0	149.50
3	ORD1004	2026-01-06	Hyderabad	Smartphone	1.0	699.00	0.0	699.00
4	ORD1005	2026-01-07	Pune	Coffee Maker	2.0	45.00	0.0	90.00

[25]:
```
df.loc[df['Revenue'] > 500, ['OrderID', 'Product', 'Revenue']]
```

[25]:

	OrderID	Product	Revenue
0	ORD1001	Laptop	899.99
3	ORD1004	Smartphone	699.00
13	ORD1016	Laptop	899.99
28	ORD1031	Laptop	899.99
31	ORD1034	Smartphone	699.00
32	ORD1034	Monitor	699.00
46	ORD1049	Smartphone	699.00
61	ORD1064	Smartphone	699.00
73	ORD1076	Wireless Mouse	899.99
76	ORD1079	Smartphone	699.00
90	ORD1094	Smartphone	699.00
99	ORD1046	Laptop	899.99
100	ORD1091	Laptop	899.99

[26]:
```
df.loc[df['DiscountPct'] > 0.10, ['OrderID', 'Product', 'DiscountPct']]
```

[26]:

	OrderID	Product	DiscountPct

[27]:
```
df.loc[df['Units'] > 2, ['OrderID', 'Product', 'Units']]
```

[27]:

	OrderID	Product	Units
5	ORD1007	Backpack	3.0
7	ORD1010	Water Bottle	4.0
9	ORD1012	Notebook Set	5.0
19	ORD1022	Backpack	3.0
22	ORD1025	Water Bottle	4.0
24	ORD1027	Notebook Set	5.0
25	ORD1028	Gaming Keyboard	3.5
29	ORD1032	Wireless Mouse	3.0
35	ORD1037	Backpack	3.0
40	ORD1042	Notebook Set	5.0
49	ORD1052	Backpack	3.0
52	ORD1055	Water Bottle	4.0

54	ORD1057	Notebook Set	5.0
58	ORD1061	Laptop	10000.0
64	ORD1067	Backpack	3.0
67	ORD1070	Water Bottle	4.0
69	ORD1072	Notebook Set	5.0
79	ORD1082	Backpack	3.0
82	ORD1085	Water Bottle	4.0
84	ORD1087	Notebook Set	5.0
93	ORD1097	Backpack	3.0
96	ORD1100	Water Bottle	4.0

[28]:
```
df.loc[(df['Region'] == 'Chennai') | (df['Product'] == 'Laptop'), ['OrderID', 'Region', 'Product', 'Revenue']]
```

[28]:

	OrderID	Region	Product	Revenue
0	ORD1001	Chennai	Laptop	899.99
6	ORD1008	Chennai	Bluetooth Speaker	59.99
8	ORD1011	Chennai	Monitor	229.99
12	ORD1015	Chennai	Air Fryer	129.99
13	ORD1016	Chennai	Laptop	899.99
20	ORD1023	Chennai	Bluetooth Speaker	59.99
22	ORD1025	Chennai	Water Bottle	48.00
23	ORD1026	Chennai	Monitor	229.99
27	ORD1030	Chennai	Air Fryer	129.99
28	ORD1031	Chennai	Laptop	899.99
36	ORD1038	Chennai	Bluetooth Speaker	59.99
39	ORD1041	Chennai	Monitor	229.99
43	ORD1045	Chennai	Air Fryer	129.99
50	ORD1053	Chennai	Bluetooth Speaker	59.99
53	ORD1056	Chennai	Monitor	229.99
57	ORD1060	Chennai	Air Fryer	129.99
58	ORD1061	Chennai	Laptop	NaN
65	ORD1068	Chennai	Bluetooth Speaker	59.99
68	ORD1071	Chennai	Monitor	229.99
72	ORD1075	Chennai	Air Fryer	129.99
73	ORD1076	Chennai	Wireless Mouse	899.99
80	ORD1083	Chennai	Bluetooth Speaker	59.99
83	ORD1086	Chennai	Monitor	29.99
87	ORD1090	Chennai	Air Fryer	129.99
94	ORD1098	Chennai	Bluetooth Speaker	59.99
95	ORD1099	Chennai	Desk Lamp	49.00
99	ORD1046	Chennai	Laptop	899.99
100	ORD1091	Chennai	Laptop	899.99

[29]:
```
df.loc[(df['Revenue'] > 100) & (df['Units'] >= 1), ['OrderID', 'Product', 'Units', 'Revenue']]
```

[29]:

	OrderID	Product	Units	Revenue
0	ORD1001	Laptop	1.0	899.990
2	ORD1003	Office Chair	1.0	149.500
3	ORD1004	Smartphone	1.0	699.000
8	ORD1011	Monitor	1.0	229.990
12	ORD1015	Air Fryer	1.0	129.990
13	ORD1016	Laptop	1.0	899.990
15	ORD1018	Office Chair	1.0	149.500
23	ORD1026	Monitor	1.0	229.990
25	ORD1028	Gaming Keyboard	3.5	314.965
27	ORD1030	Air Fryer	1.0	129.990
28	ORD1031	Laptop	1.0	899.990
30	ORD1033	<NA>	1.0	149.500
31	ORD1034	Smartphone	1.0	699.000
32	ORD1034	Monitor	1.0	699.000
39	ORD1041	Monitor	1.0	229.990
43	ORD1045	Air Fryer	1.0	129.990
45	ORD1048	Office Chair	1.0	149.500
46	ORD1049	Smartphone	1.0	699.000
53	ORD1056	Monitor	1.0	229.990
57	ORD1060	Air Fryer	1.0	129.990
60	ORD1063	Office Chair	1.0	149.500
61	ORD1064	Smartphone	1.0	699.000
68	ORD1071	Monitor	1.0	229.990
72	ORD1075	Air Fryer	1.0	129.990
73	ORD1076	Wireless Mouse	1.0	899.990
75	ORD1078	Office Chair	1.0	149.500
76	ORD1079	Smartphone	1.0	699.000
87	ORD1090	Air Fryer	1.0	129.990
89	ORD1093	Office Chair	1.0	149.500

90	ORD1094	Smartphone	1.0	699.000
99	ORD1046	Laptop	1.0	899.990
100	ORD1091	Laptop	1.0	899.990

[30]:
```
df.loc[:4, ['OrderID', 'Product', 'Revenue']]
```

[30]:

	OrderID	Product	Revenue
0	ORD1001	Laptop	899.99
1	ORD1002	Wireless Mouse	39.98
2	ORD1003	Office Chair	149.50
3	ORD1004	Smartphone	699.00
4	ORD1005	Coffee Maker	90.00

[31]:
```
df.iloc[:5, :4]
```

[31]:

	OrderID	Date	CustomerID	Product
0	ORD1001	2026-01-03	CUST001	Laptop
1	ORD1002	2026-01-04	CUST002	Wireless Mouse
2	ORD1003	2026-01-05	CUST003	Office Chair
3	ORD1004	2026-01-06	CUST004	Smartphone
4	ORD1005	2026-01-07	CUST005	Coffee Maker

[32]:
```
df.reset_index(drop=True).head()
```

[32]:

	OrderID	Date	CustomerID	Product	Category	Units	UnitPrice	ReportedAmount	PaymentMethod	OrderStatus	Region	Year	MonthNumber	Day	M
0	ORD1001	2026-01-03	CUST001	Laptop	Electronics	1.0	899.99	899.99	Credit Card	Shipped	Chennai	2026.0	1.0	3.0	
1	ORD1002	2026-01-04	CUST002	Wireless Mouse	Electronics	2.0	19.99	39.98	Upi	Delivered	Bengaluru	2026.0	1.0	4.0	
2	ORD1003	2026-01-05	CUST003	Office Chair	Furniture	1.0	149.50	149.50	Cash On Delivery	Delivered	Mumbai	2026.0	1.0	5.0	
3	ORD1004	2026-01-06	CUST004	Smartphone	Electronics	1.0	699.00	699.00	<NA>	Processing	Hyderabad	2026.0	1.0	6.0	
4	ORD1005	2026-01-07	CUST005	Coffee Maker	Home Appliances	2.0	45.00	90.00	Upi	Delivered	Pune	2026.0	1.0	7.0	

[33]:
```
region_summary = df.groupby("Region").agg(
    TotalRevenue=("Revenue", "sum"),
    Orders=("OrderID", "nunique"),
    AverageRevenuePerOrder=("Revenue", "mean")
).sort_values("TotalRevenue", ascending=False)
region_summary
```

[33]:

Region	TotalRevenue	Orders	AverageRevenuePerOrder
Chennai	7876.750	28	291.731481
Hyderabad	4893.000	7	699.000000
Mumbai	1046.500	7	149.500000
Bengaluru	1024.785	13	85.398750
Pune	540.000	7	90.000000
Kolkata	539.820	6	89.970000
Delhi	399.950	7	79.990000
Jaipur	294.000	6	49.000000
Coimbatore	270.000	6	45.000000
Ahmedabad	240.000	6	48.000000
Lucknow	124.750	6	24.950000

[34]:
```
product_summary = df.groupby("Product").agg(
    TotalRevenue=("Revenue", "sum"),
    UnitsSold=("Units", "sum"),
    Orders=("OrderID", "nunique")
).sort_values("TotalRevenue", ascending=False)
product_summary
```

[34]:

Product	TotalRevenue	UnitsSold	Orders
Laptop	4499.950	10005.0	6
Smartphone	4194.000	7.0	7
Monitor	1878.940	7.0	7
Wireless Mouse	1159.860	14.0	8
Office Chair	897.000	6.0	6
Air Fryer	779.940	6.0	6
Gaming Keyboard	764.915	8.5	6
Backpack	629.790	21.0	7
Coffee Maker	540.000	14.0	7
Bluetooth Speaker	419.930	7.0	7
Running Shoes	399.950	6.0	7
Desk Lamp	343.000	14.0	7
Water Bottle	288.000	22.0	7
Yoga Mat	270.000	12.0	6
Notebook Set	124.750	30.0	6

[35]:
```
product_summary.head(1)
```

[35]:

	TotalRevenue	UnitsSold	Orders

Product			
Laptop	4499.95	10005.0	6

[36]:
```
monthly_summary = df.dropna(subset=["Month"]).groupby("Month").agg(
    TotalRevenue=("Revenue", "sum"),
    Orders=("OrderID", "nunique")
).sort_index()
monthly_summary
```

[36]:

Month	TotalRevenue	Orders
2026-01	4652.745	27
2026-02	5006.700	27
2026-03	4350.680	31
2026-04	1400.430	11
2026-08	79.990	1

[37]:
```
monthly_summary['TotalRevenue'].idxmax()
```

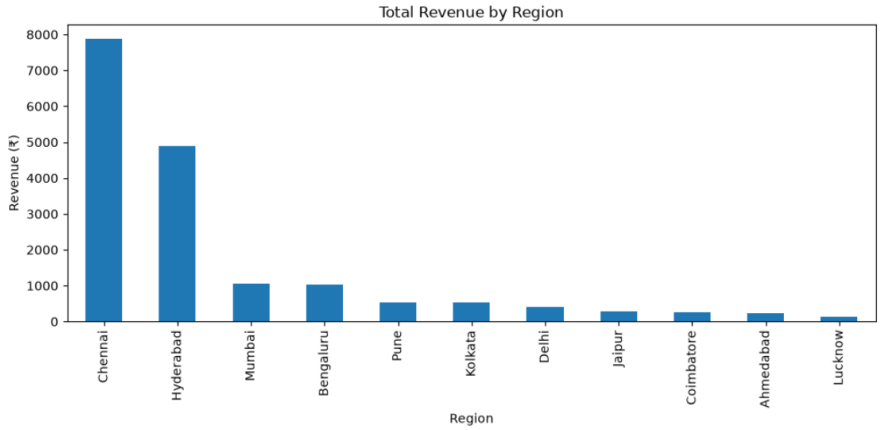
[37]:
```
'2026-02'
```

[38]:
```
highest_revenue_orders = df.sort_values("Revenue", ascending=False)[
    ["OrderID", "Date", "Region", "Product", "Units", "UnitPrice", "DiscountPct", "Revenue"]
].head(5)
highest_revenue_orders
```

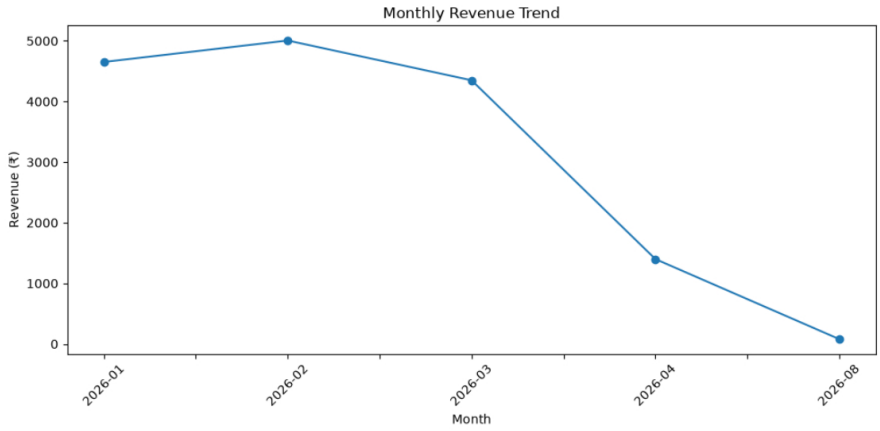
[38]:

	OrderID	Date	Region	Product	Units	UnitPrice	DiscountPct	Revenue
0	ORD1001	2026-01-03	Chennai	Laptop	1.0	899.99	0.0	899.99
13	ORD1016	2026-01-18	Chennai	Laptop	1.0	899.99	0.0	899.99
28	ORD1031	2026-02-02	Chennai	Laptop	1.0	899.99	0.0	899.99
99	ORD1046	NaT	Chennai	Laptop	1.0	899.99	0.0	899.99
100	ORD1091	NaT	Chennai	Laptop	1.0	899.99	0.0	899.99

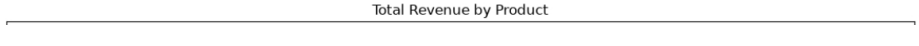
[39]:
```
region_summary["TotalRevenue"].plot(kind="bar", figsize=(10, 5))
plt.title("Total Revenue by Region")
plt.xlabel("Region")
plt.ylabel("Revenue (₹)")
plt.tight_layout()
plt.show()
```

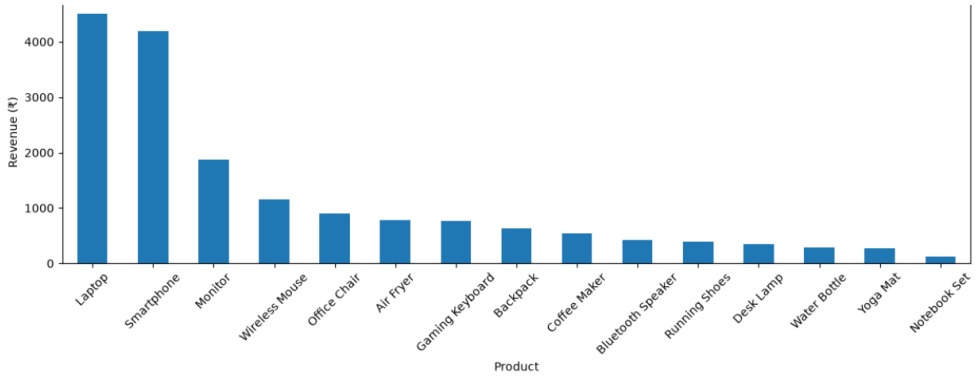


[40]:
```
monthly_summary["TotalRevenue"].plot(kind="line", marker="o", figsize=(10, 5))
plt.title("Monthly Revenue Trend")
plt.xlabel("Month")
plt.ylabel("Revenue (₹)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



[41]:
```
product_summary["TotalRevenue"].plot(kind="bar", figsize=(12, 5))
plt.title("Total Revenue by Product")
plt.xlabel("Product")
plt.ylabel("Revenue (₹)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```





[42]:
```
df['Region'].value_counts()
```

[42]:
```
Region
Chennai       28
Bengaluru     13
Hyderabad      8
Mumbai         7
Pune           7
Delhi          7
Ahmedabad      6
Lucknow        6
Coimbatore     6
Kolkata        6
Jaipur         6
Name: count, dtype: Int64
```

[43]:
```
df['Product'].unique()
```

[43]:
```
<StringArray>
[          'Laptop',    'Wireless Mouse',      'Office Chair',
      'Smartphone',       'Coffee Maker',          'Backpack',
 'Bluetooth Speaker',      'Water Bottle',           'Monitor',
     'Notebook Set',   'Gaming Keyboard',         'Yoga Mat',
        'Air Fryer',     'Running Shoes',        'Desk Lamp',
              <NA>]
Length: 16, dtype: string
```

[44]:
```
df['Revenue'].mean()
```

[44]:
```
np.float64(186.4465053763441)
```

[45]:
```
df['Revenue'].sum()
```

[45]:
```
np.float64(17339.525)
```

[46]:
```
df['Revenue'].min()
```

[46]:
```
np.float64(24.950000000000003)
```

[47]:
```
df['Revenue'].max()
```

[47]:
```
np.float64(899.99)
```

[48]:
```
df['Revenue'].median()
```

[48]:
```
np.float64(89.97)
```

[49]:
```
df['Revenue'].quantile([0.25, 0.5, 0.75])
```

[49]:
```
0.25     49.00
0.50     89.97
0.75    149.50
Name: Revenue, dtype: float64
```

[50]:
```
df.groupby(["Region", "Product"]).agg(
    TotalRevenue=("Revenue", "sum"),
    UnitsSold=("Units", "sum"),
    Orders=("OrderID", "nunique")
).reset_index().sort_values("TotalRevenue", ascending=False)
```

[50]:

	Region	Product	TotalRevenue	UnitsSold	Orders
6	Chennai	Laptop	4499.950	10005.0	6
13	Hyderabad	Smartphone	4194.000	7.0	7
7	Chennai	Monitor	1179.940	6.0	6
9	Chennai	Wireless Mouse	899.990	1.0	1
17	Mumbai	Office Chair	897.000	6.0	6
3	Chennai	Air Fryer	779.940	6.0	6
1	Bengaluru	Gaming Keyboard	764.915	8.5	6
12	Hyderabad	Monitor	699.000	1.0	1
18	Pune	Coffee Maker	540.000	14.0	7
15	Kolkata	Backpack	539.820	18.0	6
4	Chennai	Bluetooth Speaker	419.930	7.0	7
11	Delhi	Running Shoes	399.950	6.0	7
14	Jaipur	Desk Lamp	294.000	12.0	6
10	Coimbatore	Yoga Mat	270.000	12.0	6
2	Bengaluru	Wireless Mouse	259.870	13.0	7
0	Ahmedabad	Water Bottle	240.000	18.0	6
16	Lucknow	Notebook Set	124.750	30.0	6
5	Chennai	Desk Lamp	49.000	2.0	1
8	Chennai	Water Bottle	48.000	4.0	1

[51]:
```
df.groupby('Category')['Revenue'].agg(['count', 'sum', 'mean', 'min', 'max'])
```

[51]:

	count	sum	mean	min	max
Category					
Electronics	37	12687.605	342.908243	29.99	899.99

Fashion	12	1029.740	85.811667	79.99	89.97
Furniture	7	1046.500	149.500000	149.50	149.50
Home & Kitchen	6	288.000	48.000000	48.00	48.00
Home Appliances	12	1319.940	109.995000	90.00	129.99
Home Decor	7	343.000	49.000000	49.00	49.00
Sports	6	270.000	45.000000	45.00	45.00
Stationery	5	124.750	24.950000	24.95	24.95

[52]:
```
df[['OrderID', 'Revenue']].rename(columns={'Revenue': 'NetRevenue'}).head()
```

[52]:

	OrderID	NetRevenue
0	ORD1001	899.99
1	ORD1002	39.98
2	ORD1003	149.50
3	ORD1004	699.00
4	ORD1005	90.00

[53]:
```
df.drop(columns=['ReportedAmount']).head()
```

[53]:

	OrderID	Date	CustomerID	Product	Category	Units	UnitPrice	PaymentMethod	OrderStatus	Region	Year	MonthNumber	Day	Month	DiscountPct
0	ORD1001	2026-01-03	CUST001	Laptop	Electronics	1.0	899.99	Credit Card	Shipped	Chennai	2026.0	1.0	3.0	2026-01	0.0
1	ORD1002	2026-01-04	CUST002	Wireless Mouse	Electronics	2.0	19.99	Upi	Delivered	Bengaluru	2026.0	1.0	4.0	2026-01	0.0
2	ORD1003	2026-01-05	CUST003	Office Chair	Furniture	1.0	149.50	Cash On Delivery	Delivered	Mumbai	2026.0	1.0	5.0	2026-01	0.0
3	ORD1004	2026-01-06	CUST004	Smartphone	Electronics	1.0	699.00	<NA>	Processing	Hyderabad	2026.0	1.0	6.0	2026-01	0.0
4	ORD1005	2026-01-07	CUST005	Coffee Maker	Home Appliances	2.0	45.00	Upi	Delivered	Pune	2026.0	1.0	7.0	2026-01	0.0

[54]:
```
df["RevenueBand"] = df["Revenue"].apply(
    lambda x: "Unknown" if pd.isna(x) else ("High" if x >= 500 else "Regular")
)
df[["OrderID", "Revenue", "RevenueBand"]].head()
```

[54]:

	OrderID	Revenue	RevenueBand
0	ORD1001	899.99	High
1	ORD1002	39.98	Regular
2	ORD1003	149.50	Regular
3	ORD1004	699.00	High
4	ORD1005	90.00	Regular

[55]:
```
df[['Units', 'UnitPrice', 'ReportedAmount', 'DiscountPct', 'Revenue']].corr(numeric_only=True)
```

[55]:

	Units	UnitPrice	ReportedAmount	DiscountPct	Revenue
Units	1.000000	-0.009765	-0.010332	NaN	-0.369614
UnitPrice	-0.009765	1.000000	-0.010576	NaN	0.994053
ReportedAmount	-0.010332	-0.010576	1.000000	NaN	-0.057542
DiscountPct	NaN	NaN	NaN	NaN	NaN
Revenue	-0.369614	0.994053	-0.057542	NaN	1.000000

[56]:
```
quality_summary = pd.Series({
    "Rows": len(df),
    "MissingDates": df["Date"].isna().sum(),
    "MissingProducts": df["Product"].isna().sum(),
    "MissingRegions": df["Region"].isna().sum(),
    "MissingUnits": df["Units"].isna().sum(),
    "InvalidUnits": ((df["Units"] <= 0) | (df["Units"] > 100)).sum(),
    "MissingUnitPrice": df["UnitPrice"].isna().sum(),
```